

1. Zinc (65.37 g/mol) is determined by precipitating and weighing $\text{Zn}_2\text{Fe}(\text{CN})_6$ (342.70 g/mol). What mass of zinc in grams is contained in a sample that gives 0.438 g precipitate?
 - ☐ a. 2.19
 - ☐ b. 0.167
 - ☐ c. 0.334
 - ☐ d. 2.30
2. What is the concentration in ppm of a 1.5×10^{-3} molar solution of NaCl (58.5 g/mol)?
 - ☐ a. 88 ppm
 - ☐ b. 8.8 ppm
 - ☐ c. 150 ppm
 - ☐ d. 1.5 ppm
3. Which one of the following precipitates is more soluble in its saturated solution?
 - ☐ a. AgCl ($K_{\text{sp}} = 1.8 \times 10^{-10}$)
 - ☐ b. AgI ($K_{\text{sp}} = 8.5 \times 10^{-17}$)
 - ☐ c. AgBr ($K_{\text{sp}} = 5.0 \times 10^{-13}$)
 - ☐ d. Ag_2CrO_4 ($K_{\text{sp}} = 1.2 \times 10^{-12}$)
4. Given that the solubility product for $\text{La}(\text{IO}_3)_3$ is 1.0×10^{-11} , what is the concentration of La^{3+} in a saturated solution of lanthanum iodate?
 - ☐ a. 1.35×10^{-3} M
 - ☐ b. 1.0×10^{-3} M
 - ☐ c. 7.18×10^{-5} M
 - ☐ d. 7.8×10^{-4} M
5. How many grams of 2.50% by mass aqueous HF (20.0 g/mol) are required to provide 20% excess to react with 50.0 mL of 0.0325 M Th^{4+} by the reaction:
 $\text{Th}^{4+} + 4 \text{F}^- \rightarrow \text{ThF}_4 (\text{s})$
 - ☐ a. 3.12
 - ☐ b. 25.0
 - ☐ c. 6.24
 - ☐ d. 12.5

6. At 700 K, the reaction $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ has the equilibrium constant $K_c = 4.3 \times 10^6$, and the following concentrations are present: $[\text{SO}_2] = 0.10 \text{ M}$; $[\text{SO}_3] = 10. \text{ M}$; $[\text{O}_2] = 0.10 \text{ M}$. Is the mixture at equilibrium? If not, in which direction (as the equation is written), left to right or right to left, will the reaction proceed to reach equilibrium?
- ☐ a. There is not enough information to predict the direction
 - ☐ b. No, right to left
 - ☐ c. No, left to right
 - ☐ d. Yes, the mixture is at equilibrium
7. A student performs an experiment to determine the density of a sugar solution. She obtains the following results: 4.71 g/mL, 4.73 g/mL, 4.67 g/mL, 4.69 g/mL. If the actual value for the density of the sugar solution is 4.40 g/mL, which statement below best describes her results?
- ☐ a. Her results are both precise and accurate
 - ☐ b. Her results are precise, but not accurate
 - ☐ c. Her results are accurate, but not precise
 - ☐ d. It isn't possible to determine with the information given
8. How many milliliters of 0.260 M Na_2S are needed to react with 40.00 mL of 0.315 M AgNO_3 ?
 $\text{Na}_2\text{S}(\text{aq}) + 2 \text{AgNO}_3(\text{aq}) \rightarrow 2 \text{NaNO}_3(\text{aq}) + \text{Ag}_2\text{S}(\text{s})$
- ☐ a. 12.1 mL
 - ☐ b. 66.0 mL
 - ☐ c. 24.2 mL
 - ☐ d. 48.5 mL
9. Determination of the sodium level in separate portions of a blood sample by ion-selective electrode measurement gave the following results: 139.2, 139.8, 140.1, and 139.4 meq/L. What is the range within which the true value falls, assuming no determinate error at the 99% confidence level (t at 99% confidence level = 5.84)?
- ☐ a. 139.6 ± 0.6
 - ☐ b. 139.6 ± 0.5
 - ☐ c. 139.6 ± 1.5
 - ☐ d. 139.6 ± 1.2
10. What concentration of Ag^+ is required to precipitate ONLY AgBr in a solution that contains both Br^- and Cl^- at a concentration of 0.040 M?
 K_{sp} of $\text{AgCl} = 1.6 \times 10^{-10}$ and K_{sp} of $\text{AgBr} = 7.7 \times 10^{-13}$

- a. $1.0 \times 10^{-11} \text{ M} < [\text{Ag}^+] < 2.0 \times 10^{-9} \text{ M}$ $1.0 \times 10^{-11} \text{ M} < [\text{Ag}^+] < 2.0 \times 10^{-9} \text{ M}$
- b. $1.9 \times 10^{-11} \text{ M} < [\text{Ag}^+] < 4.0 \times 10^{-9} \text{ M}$ $1.9 \times 10^{-11} \text{ M} < [\text{Ag}^+] < 4.0 \times 10^{-9} \text{ M}$
- c. $1.0 \times 10^{-11} \text{ M} < [\text{Ag}^+] < 2.0 \times 10^{-9} \text{ M}$ $1.0 \times 10^{-11} \text{ M} < [\text{Ag}^+] < 2.0 \times 10^{-9} \text{ M}$
- d. $3.9 \times 10^{-11} \text{ M} < [\text{Ag}^+] < 8.0 \times 10^{-9} \text{ M}$ $3.9 \times 10^{-11} \text{ M} < [\text{Ag}^+] < 8.0 \times 10^{-9} \text{ M}$

11. Which of the following statements is most correct?

- a. Disturbance of an equilibrium system results in a shift of the equilibrium position
- b. Adding more reactants for a reaction at equilibrium decreases the value of K
- c. Always salts with the same value of K_{sp} have similar solubility
- d. At equilibrium, no more reactants are transformed into products

12. How many significant figures are there in the answer for the following calculation?

$$(143.7 - 121) \times 2.060.6000.600(143.7 - 121) \times 2.06$$

- a. Three
- b. Four
- c. Two
- d. One

13. The following replicates are obtained for the analysis of calcium in drinking water: 23.4, 23.2, 22.8, 26.0, 24.0, and 23.0 ppm. After applying the Grubbs Test, the mean is --- and the standard deviation is ---. Use $G_{table} = 1.822$.

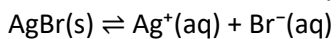
- a. 23.7 and 1.2
- b. 23.7 and 0.5
- c. 23.3 and 1.2
- d. 23.3 and 0.5

14. What is the molar solubility of Hg_2Cl_2 ($K_{sp} = 1.2 \times 10^{-18}$) in 0.060 M KCl?

- a. $2.6 \times 10^{-15} \text{ M}$
- b. $1.3 \times 10^{-15} \text{ M}$
- c. $6.6 \times 10^{-16} \text{ M}$
- d. $3.3 \times 10^{-16} \text{ M}$

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
C	A	D	A	C	C	B	C	A	B	A	A	A	D

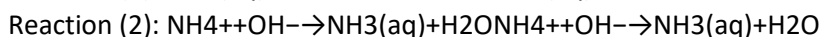
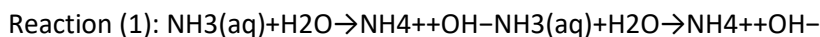
15. What is the molar solubility of AgBr in (a) pure water?



$$K_{sp} = 7.7 \times 10^{-13}$$

- ☐ a. 8.0×10^{-7}
- ☐ b. 9.2×10^{-8}
- ☐ c. 5.7×10^{-8}
- ☐ d. 8.8×10^{-7} M

16. What is the equilibrium constant for reaction (2), if the equilibrium constant for reaction (1) is 1.8×10^{-5} ?

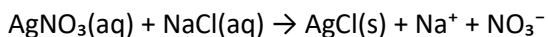


- ☐ a. 5.6×10^{-1}
- ☐ b. 5.6×10^4
- ☐ c. 5.6×10^{-4}
- ☐ d. 5.6×10^{-1}

17. The ionic strength of a solution containing 0.1 M FeCl_3 , 0.2 M KCl , and 1 M NH_3 is:

- ☐ a. 0.9
- ☐ b. 0.4
- ☐ c. 0.8
- ☐ d. 0.6

18. What volume of 25.0 wt% AgNO_3 (170 g/mol, density = 1.4 g/mL) is required to precipitate all chloride Cl^- (35.5 g/mol) in a 100 mL 0.25 M solution and give a 5% excess?



- ☐ a. 14.5 mL
- ☐ b. 12.7 mL
- ☐ c. 12.5 mL
- ☐ d. 12.1 mL

19. One part per million (ppm) is equivalent to one:

- ☐ a. mg/g
- ☐ b. ng/g
- ☐ c. $\mu\text{g/g}$

- d. $\mu\text{g/kg}$
- 20. The following replicates are obtained for the analysis of calcium in drinking water: 23.4, 23.2, 22.8, 26.0, 24.0, and 23.0 ppm. After applying the Grubbs Test, the mean is --- and the standard deviation is ---. Use $G_{\text{table}}=1.822$.
- a. 23.3 and 0.5
 - b. 23.3 and 1.2
 - c. 23.7 and 1.2
 - d. 23.7 and 0.5
- 21. Choose the correct answer with the correct significant figures:
 $(12)+(3.2)-(1.24)=?$
- a. 14.0
 - b. 13.96
 - c. 14.16
 - d. 14
- 22. The measured pH of 0.0100 M HCl + 0.0900 M KCl at 25°C is 2.102. From this information, calculate the activity coefficient of H^+ in this solution.
- a. 0.791
 - b. 1.00
 - c. 1.05
 - d. 0.890
- 23. You have 505 mL of 0.150 M HCl solution, and you want to dilute it to exactly 0.100 M. How much water would you add?
- a. 1009 mL
 - b. 126 mL
 - c. 757 mL
 - d. 252 mL
- 24. The wt% and the ppm concentration of a solution made by dissolving 10.0 g salt in 240 g deionized water (density of water = 1.0 g/mL) are:
- a. 4%, 40000 ppm
 - b. 2%, 2000 ppm
 - c. 1.66%, 16600 ppm

- d. 4%, 4000 ppm

25. Convert 5.5 Tm (terameter) to km:

- a. 5.5×10^{15}
- b. 5.5×10^9
- c. 5.5×10^{12}
- d. 5.5×10^6

Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25
D	B	A	B	C	A	A	A	C	C	B

26. The pH ($-\log[H^+]$) of a solution is 4.164. The $[H^+]$ is:

- a. 6.85×10^{-5} M
- b. 7.0×10^{-5} M
- c. 6.8×10^{-5} M
- d. 5.850×10^{-5} M

27. The density of a 5.24 M $NaHCO_3$ (84.0 g/mol) solution is 1.19 g/mL. Its molality is:

- a. 6.99
- b. 7.27
- c. 9.97
- d. 6.71

28. Suppose that $MgBr_2$ dissolves in pure water to give Mg^{2+} , Br^- , and $MgBr^+$. The mass balance for Br^- in 0.15 M aqueous solution of $MgBr_2$ is:

- a. $0.45 \text{ M} = 2[Br^-] + [MgBr^+]$
- b. $0.30 \text{ M} = [Br^-] + [MgBr^+]$
- c. $0.45 \text{ M} = [Br^-] + [MgBr^+]$
- d. $0.30 \text{ M} = 2[Br^-] + [MgBr^+]$

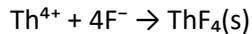
29. How many grams of $CuSO_4 \cdot 5H_2O$ (249.68 g/mol) should be dissolved in 250 mL to make 4.00 mM Cu^{2+} ?

- a. 0.5
- b. 0.999
- c. 0.249
- d. 0.499

30. $[\log \frac{1}{3.4 \times 10^{-4}}] + 4.45 = ?$ $[\log 3.4 \times 10^{-4}] + 4.45 = ?$

- a. 1.00
- b. 0.98
- c. 0.95
- d. 0.99

31. How many grams of 0.491 wt% aqueous HF (20.0 g/mol) are required to provide a 50% excess to react with 25.0 mL of 0.0236 M Th^{4+} by the reaction:



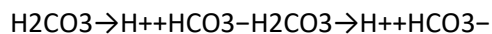
- a. 1.8
- b. 9.61
- c. 14.4
- d. 7.2

32. The $[\text{H}^+]$ of a solution is 2.70×10^{-6} . What is the pH of the solution?

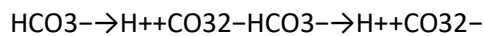
- a. 5.5686
- b. 5.57
- c. 5.569
- d. 44.48

33. What is the pH at which $[\text{CO}_3^{2-}] = 2[\text{HCO}_3^-]$?

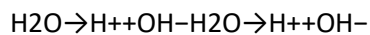
Given:



$$K_{a1} = 3.0 \times 10^{-4} \quad K_{a1} = 3.0 \times 10^{-4}$$

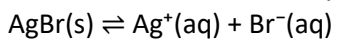


$$K_{a2} = 2.2 \times 10^{-7} \quad K_{a2} = 2.2 \times 10^{-7}$$



- a. 5.9
- b. 6.96
- c. 4.4
- d. 7.8

34. What is the molar solubility of AgBr in 0.0010 M NaBr?



$$K_{sp}=7.7 \times 10^{-13}$$

- ☐ a. 8.8×10^{-7}
- ☐ b. 3.5×10^{-7}
- ☐ c. 7.7×10^{-10}
- ☐ d. 3.578×10^{-7}

35. The following data are for the determination of lead in gasoline: 5.2, 6.8, 6.5, 6.3, and 5.9. What is the true value at 90% confidence level? (t-table at 90% CL = 2.132).

- ☐ a. 6.2 ± 0.59
- ☐ b. 6.1 ± 0.66
- ☐ c. 6.14 ± 0.59
- ☐ d. 6.1 ± 0.56

36. Choose the correct statement:

- ☐ a. Molarity is temperature-dependent
- ☐ b. Molarity is always higher than molality
- ☐ c. Molality is temperature-dependent
- ☐ d. Molality is always higher than molarity

Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36
A	B	A	B	B	B	B	B	C	A	A